

1	When a current I is passed through the length of a metal plate of thickness d in a uniform magnetic field with a magnetic flux density B, a potential difference V is generated across its width. This potential difference, known as the Hall Voltage, is given by the relationship $V = \frac{BI}{ned}$ where n is a physical quantity related to the type of metal used, and e is the charge of the electron.	
	(a) By using the definition of the magnetic flux density, show that the SI base units of B is $\text{kg s}^{-2} \text{A}^{-1}$.	[2]
	(a) Solution: Definition of magnetic flux density: $B = \frac{F}{IL}$ $[B] = \frac{[F]}{[I][L]} = \frac{\text{kg m s}^{-2}}{\text{A m}}$ $[B] = \text{kg s}^{-2} \text{A}^{-1}$	1 1
	(b) In practice, the Hall Voltage is used to measure the value of the magnetic flux density in a region.	
	(i) In a particular experiment, a silver plate is placed in a uniform magnetic field of an unknown magnetic flux density B. The following are the recorded values of the various quantities along with their uncertainties. $V = (7.9 \pm 0.1) \mu\text{V}$ $I = (100 \pm 1) \text{A}$ $d = (0.20 \pm 0.01) \text{cm}$ The value of n for silver is $5.9 \times 10^{28} \text{ m}^{-3}$ and this value is known to a high degree of precision. Determine the magnitude of B with its associated uncertainty.	

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Fig. 2.1 shows a 0.50 kg trolley placed at the top of a smooth ramp.